# Species-Aware Module Review: Choline uptake and glycine betaine biosynthesis (UPA00529) in *Pseudomonas putida* KT2440

**Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module / bucket:** unipathway:UPA00529 — "Amine and polyamine biosynthesis; betaine biosynthesis via choline pathway" **Commissioned candidate genes:** betI / PP_5719 (regulator only)

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## 1. Executive summary

The choline → betaine-aldehyde → glycine-betaine module is **fully satisfiable** in *P. putida* KT2440. All three functional steps are encoded by direct, high-confidence orthologs that sit in a single genomic cluster at ~5.77 Mb, together with the cognate regulator:

- **Choline uptake:** BetT-I / **PP_5061** (BCCT-family choline transporter), assigned by decisive genomic context (co-clustered with *betBA*); two BCCT paralogs (PP_0229, PP_3957) belong to unrelated operons and are excluded.
- **Choline → betaine aldehyde:** BetA / **PP_5064** (choline dehydrogenase, EC 1.1.99.1; HAMAP MF_00750; **77.7%** identity to *E. coli* BetA).
- **Betaine aldehyde → glycine betaine:** BetB / **PP_5063** (betaine-aldehyde dehydrogenase, EC 1.2.1.8; HAMAP MF_00804; **77.1%** identity to *E. coli* BetB).
- **Regulation:** BetI / **PP_5719** (TetR/HTH choline-responsive repressor), physically within the cluster between *betT-I* and *betB*.

The single most important curation issue is that the **local metadata bucket captured only the regulator (betI), not the catalytic core.** The enzymes that actually carry the UniPathway annotation (betA/PP_5064, betB/PP_5063) and the transporter (betT-I/PP_5061) were absent from the candidate list. The module should be marked **covered**, but the bucket **needs revision** to add and promote the catalytic genes.

Direct genetic evidence in KT2440 (Galvão et al. 2006, P 17116241) confirms *betBA* are required for choline→glycine-betaine conversion and osmotolerance, and — importantly — that the neighboring *betC* (choline-*O*-sulphate) function is **uncoupled from osmoprotection** and should be kept out of this module.

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## 2. Target-organism pathway definition

**Included process:** transmembrane import of exogenous choline, followed by its two-step oxidation to the compatible solute glycine betaine:

```
choline (out) --BetT--> choline (in) --BetA(EC 1.1.99.1)--> betaine aldehyde --BetB(EC 1.2.1.8)
```

This is the canonical bacterial "betaine biosynthesis via choline pathway" (UniPathway **UPA00529**; equivalent to MetaCyc *choline degradation I* / KEGG map00260 betaine node). Glycine betaine then acts as an osmoprotectant (accumulated under high salt) and, in *P. putida*, also as a carbon/nitrogen source.

**Neighboring pathways to keep separate:** - **Choline-*O*-sulphate utilization** (BetC / PP_0077 choline-sulfatase, EC 3.1.6.6, + ABC transporter/LysR regulator at PP_0076–0077). Galvão 2006 shows this is genetically distant from *betBA* and functionally dedicated to COS catabolism, **not** osmoprotection. - **Direct glycine-betaine/quaternary-amine transport** (ProU/OpuA-type ABC importers, ProP): these supply betaine without synthesis and belong to an osmolyte-transport module, not the biosynthetic one. - **Downstream glycine betaine catabolism** (demethylation to dimethylglycine) is a separate degradation pathway.

**Alternate names / DB definitions:** *bet* regulon; choline oxidation pathway; UniProt pathway string "Amine and polyamine biosynthesis; betaine biosynthesis via choline pathway"; MetaCyc PWY-... choline degradation.

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## 3. Expected step model

| Step | Function (GO/EC) | Expected player | KT2440 status |
|---|---|---|---|
| 1. Choline uptake | choline transmembrane transporter (GO:0015220) | BetT-family (BCCT) | **Covered** — BetT-I/PP_5061 (cluster); paralogs PP_0229, PP_3957 |
| 2. Choline oxidation | choline dehydrogenase (EC 1.1.99.1) | FAD/GMC BetA | **Covered** — BetA/PP_5064 (cluster); paralog PP_0056 |
| 3. Betaine aldehyde oxidation | betaine-aldehyde dehydrogenase, NAD+ (EC 1.2.1.8) | BetB (ALDH) | **Covered** — BetB/PP_5063 (cluster) |
| (Regulation) | TetR repressor, choline-responsive | BetI | Present — BetI/PP_5719 (cluster) |

## 4. Candidate genes and evidence

**Genomic layout (KEGG coordinates, ~5.77 Mb) — the *bet* cluster:**

```
PP_5061 betT-I  5,770,959–5,772,962 (–)   choline transporter (BCCT)
PP_5719 betI    5,773,438–5,774,094 (+)   TetR repressor
PP_5063 betB    5,774,134–5,775,606 (+)   betaine-aldehyde dehydrogenase
PP_5064 betA    5,775,684–5,777,381 (+)   choline dehydrogenase
```

> **Curation note on locus tags:** despite its high number, **PP_5719 (betI) physically lies between betT-I and betB** — it is the cognate, adjacent regulator of the *betBA* operon. The layout recapitulates the canonical *E. coli bet* regulon, where *betI* and three structural genes *betT* (choline porter), *betA* (choline dehydrogenase) and *betB* (betaine-aldehyde dehydrogenase) are controlled by two **divergent promoters** driving *betT* vs *betIBA*, with **BetI acting as a choline-responsive repressor of both** (Lamark et al. 1996, ;
> P 8626294
> gene order established by Andresen et al. 1988,

> ).
> P 3065456
> In KT2440 *betT-I*/PP_5061 is on the **reverse strand** while *betI-betB-betA* are on the **forward strand** — exactly the divergent arrangement expected. Species transfer of this regulon model to KT2440 is **strong**. The high locus tag on betI is a re-annotation artifact, not evidence of a distal/unrelated gene.

**Orthology quantification (global-alignment % identity to *E. coli* K-12 reference):**

| KT2440 protein | vs *E. coli* ortholog | Paralog vs same ref | Verdict |
|---|---|---|---|
| BetA / PP_5064 | **77.7%** vs EcBetA (P17444) | PP_0056 = 40.5% | PP_5064 = true ortholog; PP_0056 distant paralog |
| BetB / PP_5063 | **77.1%** vs EcBetB (P17445) | PP_0708 = 40.7% | PP_5063 = true ortholog; PP_0708 distant paralog |
| BetT-I / PP_5061 | 43.4% vs EcBetT (P0ABC9) | PP_0229 = 46.2%; PP_3957 = 46.5% | No strong sequence ortholog; PP_5061 chosen on **genomic context**, transporter step ambiguous |

> Interpretation: the two catalytic steps are backed by unambiguous ~77% orthologs, so PP_5064 and PP_5063 are safe module representatives and PP_0056/PP_0708 are correctly demoted to candidate_uncertain. The three BCCT transporters are all only ~43–46% identical to *E. coli* BetT (and 45–51% to each other), so **step 1 cannot be resolved by sequence alone** — PP_5061 is preferred only because it co-clusters with *betIBA*; PP_0229/PP_3957 remain plausible choline importers.

| Gene | Accession | Role | Evidence type | Curation caveat |
|---|---|---|---|---|
| **betA / PP_5064** | Q88CW6 | Choline dehydrogenase (step 2) | UniProt HAMAP MF_00750; RHEA:17433/15305; **direct KT2440 mutant phenotype** (Galvão 2006) | Dual EC (1.1.99.1 + 1.2.1.8) is **genuine bifunctionality** — BetA oxidizes both choline and betaine aldehyde (Cánovas 2000, P 10708384), not an annotation error; BetB remains the dedicated NAD+-BADH. **Promote.** |
| **betB / PP_5063** | Q88CW7 | Betaine-aldehyde dehydrogenase (step 3) | UniProt HAMAP MF_00804; RHEA:15305; direct KT2440 phenotype (Galvão 2006) | High-confidence ortholog in cluster. **Promote.** |
| **betT-I / PP_5061** | Q88CW9 | Choline transporter (step 1) | Cluster position, BCCT family; BetT is the osmoregulated choline importer (cryo-EM mechanism, Yang 2024, P 39141726) | No EC/GO in UniProt; transport not directly assayed in KT2440. **Promote as best transporter candidate.** |
| **betI / PP_5719** | A0A140FWS5 | Choline-responsive TetR repressor | HAMAP MF_00768; cluster position | Regulatory, **not** a catalytic step; should not itself "satisfy" any enzymatic module step. |

## 5. Gaps, ambiguities, and likely over-annotations

- **No true gaps.** All three enzymatic/transport steps have strong candidates; the module is not missing any step in KT2440.

- **Transporter paralog ambiguity — resolved by genomic context:** three BCCT choline/betaine transporters exist, but their operonic contexts differ decisively (KEGG neighbor analysis):
- **BetT-I/PP_5061** — directly flanked by *betB*/PP_5063 and *betA*/PP_5064 → **the module choline importer** (step 1, covered).
- **BetT-II/PP_0229** — embedded among sulfur-assimilation genes (sulfur-compound ABC transporter PP_0226, cystine-binding PP_0227, sulfonate dioxygenase PP_0230, taurine ABC transporter PP_0231–0232) → unrelated physiology → **candidate_uncertain, not the module transporter**.
- **BetT-III(BetTC)/PP_3957** — next to NhaA $Na^+/H^+$ antiporter (PP_3958), a chloride channel (PP_3959) and its own TetR regulator (PP_3960) → a standalone osmoregulated transporter → **candidate_uncertain, not the module transporter**. Additional ABC importers (cbcVWX/PP_0294–0296, betX/PP_1741, opuA/PP_2774, yehWXYZ/PP_0868–0871, proP/PP_2914, PP_0076 betaine-binding) mostly transport betaine/carnitine rather than feed choline oxidation → not module step 1.
- **Choline-dehydrogenase paralog: BetA-I / PP_0056** (Q88RS3, EC 1.1.99.1, 550 aa) is a second choline dehydrogenase outside the cluster; likely accessory or over-propagated for this module → **candidate_uncertain**.
- **BADH-family over-count:** PP_0708 is annotated "betaine-aldehyde dehydrogenase" without EC, and >10 other ALDH-family paralogs exist. Only **BetB/PP_5063** carries the pathway/HAMAP evidence → others are **likely over-annotations** for this step.
- **BetA dual EC (1.1.99.1 + 1.2.1.8):** NOT an over-annotation — BetA-family flavoenzymes genuinely oxidize both choline and betaine aldehyde (Cánovas 2000, P 10708384 ). The dedicated NAD+-dependent step 3, however, is BetB's role; retain both enzymes.
- **BetC/PP_0077 (choline-sulfatase):** a distinct pathway (choline-*O*-sulphate → choline + sulfate). **Keep out of UPA00529.** Direct KT2440 evidence: it is downregulated at high salt and dispensable for osmoprotection (Galvão 2006).

## 6. Module and GO-curation recommendations

| Module step | Recommended status | Representative gene |
|---|---|---|
| Choline uptake | **covered** — BetT-I/PP_5061 assigned by decisive genomic context (co-clustered with betBA); paralogs PP_0229/ PP_3957 excluded (sulfur- and ion-stress operons) | betT-I / PP_5061 |
| Choline → betaine aldehyde | **covered** | betA / PP_5064 |
| Betaine aldehyde → glycine betaine | **covered** | betB / PP_5063 |
| (Regulation) | present, non-catalytic | betI / PP_5719 |

- **module_needs_revision (metadata):** the bucket must be expanded from betI-only to include betA/PP_5064, betB/PP_5063, betT-I/PP_5061. Recording only the regulator would falsely imply the pathway is regulator-defined.
- **Generic module boundaries:** correct for KT2440 — keep the three-step scope; explicitly **exclude** choline-*O*-sulphate (betC) and betaine ABC-import as separate modules.
- **GO curation:** BetT-I/PP_5061 lacks a GO transporter term in UniProt — a **GO annotation (choline transmembrane transporter activity, GO:0015220) request** is warranted based on family + cluster context. Flag the BetA dual-EC for review.
- No new module document is required; the existing generic outline maps cleanly onto the KT2440 cluster.

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## 7. Genes to promote to full `fetch-gene` review

1. **betA / PP_5064** (Q88CW6) — choline dehydrogenase; resolve dual-EC caveat.
2. **betB / PP_5063** (Q88CW7) — betaine-aldehyde dehydrogenase; high-confidence step 3.
3. **betT-I / PP_5061** (Q88CW9) — primary choline uptake candidate; needs transport-evidence review.
4. (Context) **betI / PP_5719** — confirm regulator, keep as regulatory annotation only.

5. (Disambiguation) **PP_0056, PP_0229, PP_3957, PP_0708** — review to confirm they are paralogs/other-context, not module representatives.

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## 8. Key references


- Galvão TC, de Lorenzo V, Cánovas D. (2006) *Uncoupling of choline-O-sulphate utilization from osmoprotection in Pseudomonas putida.* Mol Microbiol.

  P 17116241

  — Direct KT2440 evidence that *betBA* convert choline to glycine betaine and confer osmotolerance, and that *betC* (COS) is separate from osmoprotection.
- Lamark T, Røkenes TP, McDougall J, Strøm AR. (1996) *The complex bet promoters of Escherichia coli: regulation by oxygen (ArcA), choline (BetI), and osmotic stress.* J Bacteriol.

  P 8626294

  — Defines the *betI*/*betT*/*betA*/*betB* regulon and BetI as a choline-responsive repressor of divergent promoters (model transferred to KT2440).
- Andresen PA et al. (1988) *Molecular cloning, physical mapping and expression of the bet genes... of Escherichia coli.* Mol Microbiol.

  P 3065456

  — Original definition of *betA/betB/betT* activities and gene order.
- Cánovas D et al. (2000) *Genes for the synthesis of the osmoprotectant glycine betaine from choline in Halomonas elongata DSM 3043.*

  P 10708384

  — *betIBA* operon; evidence that BetA oxidizes both choline and betaine aldehyde (supports BetA/PP_5064 dual EC).
- Yang et al. (2024) *Structure and mechanism of the osmoregulated choline transporter BetT.*

  P 39141726

  — Cryo-EM mechanism confirming BetT as the osmoregulated choline importer feeding betaine synthesis.

- UniProt Knowledgebase entries: Q88CW6 (BetA/PP_5064), Q88CW7 (BetB/PP_5063), Q88CW9 (BetT-I/PP_5061), A0A140FWS5 (BetI/PP_5719); HAMAP rules MF_00750, MF_00804, MF_00768.
- KEGG GENES (ppu) locus records for PP_5061–5065 and PP_5719 (genomic coordinates / operon layout).

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## Uncertainty & species-transfer notes

- Steps 2–3 (betA/betB) are **direct experimental** for KT2440 (mutant phenotypes, Galvão 2006).
- Step 1 (betT-I) is **inferred from family** + **cluster position**; transport not directly assayed in KT2440 — transfer confidence moderate–strong.
- Paralog assignments and the BetA dual-EC rest on homology/annotation and warrant expert confirmation.

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